Software validation report

Record of the validation of the data-processing software before use and after changes (ISO/IEC 17025:2017 7.11.2). Every reference example is recomputed from its inputs and compared with the published values within a documented tolerance.

Software	KAPSAM
Version	0.1.0 (commit 7269a1eeab81, 2026-11-09)
Compiler	rustc 1.97.0 (2d8144b78 2026-07-07)
Platform	linux x86_64
Date	2026-11-10
Options	Monte Carlo evaluations included (fixed seeds and numbers of trials as stored in the files)
Digest of the checked files	d7b99de18c3f673c52e94d7823ad2ca5ab4df8cb669dcb6f1181536fa99639d4 (SHA-256 of the sorted list "SHA-256 file" of all checked files; each file holds its inputs, reference values and settings (seeds, numbers of trials, rounding))

Summary

286 project files and 1865 compared values. Passed: 286; failed: 0; without reference values: 0.

All reference values are reproduced within their tolerances.

Results by method

Method	Projects	Values	Outcome
top-down	39	165	passed
GUM/MCM	134	517	passed
JCGM 106	30	252	passed
ISO 6143	8	87	passed
ISO 13528	9	183	passed
JCGM 102	17	216	passed
ISO 5725-2	6	163	passed
Nordtest TR 569	22	98	passed
Eurachem UfS:2019	7	61	passed
EURAMET cg-18 App. C	9	103	passed
GUM/MCM + JCGM 106	5	20	passed

Results by project: reference examples

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
bettencourt2026-fluoranthene-aniline	R. J. N. Bettencourt da Silva, Section 3.2; Supplementary Material 06, sheets Trueness, Performance Model and Analysis sequence	13	0.049	passed	c61c1a40642b
bettencourt2026-fluoranthene-weighted	R. J. N. Bettencourt da Silva, Section 3.2; Supplementary Material 06, sheets Trueness, Performance Model and Analysis sequence	13	0.048	passed	3deea1ea5f1e
bipm-isrd-01-qnmr-dmso2-ma-d20	BIPM, Table 4	2	0.68	passed	12f2b939fe63
bipm-isrd-01-qnmr-dmtp-ma-dmso2-d6	BIPM, Table 6	2	0.86	passed	0bc8eda567b9
bipm-isrd-02-qnmr-dmtp-dmso2-d6-M3	BIPM, Table 6	2	0.23	passed	d850f5e5e799
bipm-isrd-02-qnmr-ma-dmso2-d20	BIPM, Table 4	2	0.38	passed	c8ab925f8132
bipm-isrd-06-qnmr-ba-btmsb-d4-dmso2	BIPM, Table 6	9	0.33	passed	5007511866b4

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
bipm-isrd-06-qnmr-dmso2-btmsb-d4-dss	BIPM, Table 6	2	0.24	passed	b0fe9a6370b2
bipm-isrd-07-qnmr-dmso2-dss-d6-tsp	BIPM, Table 4	2	0.75	passed	e3960bfa4552
bipm-isrd-07-qnmr-ma-dss-d6-tsp	BIPM, Table 6	2	0.44	passed	329b28faa3a2
bipm-isrd-08-qnmr-btmsb-d4-bapdc	BIPM, Table 2	2	0.31	passed	e9b31c2d4d82
emue-e2.1-multicomponent-mediation	EMPIR 17NRM05 EMUE, Example E2.1, Tables E2.1.1–E2.1.2, E2.1.6	7	0.45	passed	eb481dbe7beb
emue-e3.3-no2-analyser-dynamic-linear	EMPIR 17NRM05 EMUE, Example E3.3, Tables E3.3.2–E3.3.5, (E3.3.17)–(E3.3.19)	4	0.38	passed	bd893c0d6eaa
emue-e3.3-no2-analyser-dynamic-nonlinear	EMPIR 17NRM05 EMUE, Example E3.3, Tables E3.3.2–E3.3.5, (E3.3.17)–(E3.3.19)	7	0.38	passed	bcd85ed3a332
emue-e3.4-soil-lead-iso21748-mandible	EMPIR 17NRM05 EMUE, E3.4.6–E3.4.7, Tables E3.4.1–E3.4.7, equation (E3.4.18)	3	0.30	passed	d1725afa02ae
emue-e3.4-soil-lead-iso21748	EMPIR 17NRM05 EMUE, E3.4.6–E3.4.7, Tables E3.4.1–E3.4.7, equation (E3.4.18)	3	0.46	passed	0442de5e853f
emue-e3.8-ammonia-permeation	EMPIR 17NRM05 EMUE, Example E3.8, (E3.8.1)–(E3.8.4), Tables E3.8.3–E3.8.5	5	0.11	passed	a75bd9bb7046
emue-e6.5-ephedrine-wada-decision-limit	EMPIR 17NRM05 EMUE, Example E6.5, E6.5.2–E6.5.3, Table E6.5.2	4	0.27	passed	a6212175abc9
emue-e6.5-ephedrine-wada-topdown	EMPIR 17NRM05 EMUE, Example E6.5, Table E6.5.1	3	0.42	passed	47870398395e
eurachem-compliance-b1-nickel-steel	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 1; Annex A, Case 1b	4	0.45	passed	bbea091120fb
eurachem-compliance-b2-upper-limit	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 2; Annex A, Case 2	3	0.20	passed	e98c6dc46045
eurachem-compliance-b3-banned-substance-exact	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.43	passed	8c3fcae894df
eurachem-compliance-b3-banned-substance-mc	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.46	passed	a691bc80be45
eurachem-compliance-b3-banned-substance-taylor	Eurachem/CITAC Guide: Use of uncertainty information in compliance assessment, Annex B, Example 3; Annex A, Case 4 and Table 1	3	0.45	passed	122a58cace7f
gum5-e1-ph-temperature-interpolation-mc	JCGM GUM-5:2026, Example 1, 1.6, 1.9 and 1.10.3	4	0.95	passed	f00bd91918ec
gum5-e1-ph-temperature-interpolation	JCGM GUM-5:2026, Example 1, 1.6, 1.9 and 1.10.3	4	0.95	passed	bd687dafbcc6
gum5-e1-ph-two-point-25c-correlated	JCGM GUM-5:2026, Example 1, 1.10.3.4 and 1.10.3.5	2	0.65	passed	399dfe74085e
gum5-e1-ph-two-point-25c	JCGM GUM-5:2026, Example 1, 1.8	4	0.55	passed	a46f9dc42415
gum5-e10-tspm-global-risks-accidental	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.17	passed	196e2ebbf055
gum5-e10-tspm-global-risks-accidental-mc	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.19	passed	a5e6abf8a3d3
gum5-e10-tspm-global-risks-shared-risk	JCGM GUM-5:2026, Example 10, 10.6.3	3	0.14	passed	ba9d50ed7edc
gum5-e10-tspm-specific-risks	JCGM GUM-5:2026, Example 10, 10.6.4 and 10.7.2	4	0.18	passed	557f0ccf9176
gum5-e12-between-bottle-homogeneity	JCGM GUM-5:2026, Example 12, Tables 12.1–12.3, 12.4.5 and 12.4.8	14	0.88	passed	9678e5ee9175
gum5-e2-benzo-a-pyrene-ensemble	JCGM GUM-5:2026, Example 2, Tables 2.1–2.3; Willink, Metrologia 44 (2007) 340, §4.1.1	3	0.57	passed	12a36c83cdc4
gum5-e2-benzo-a-pyrene	JCGM GUM-5:2026, Example 2, Tables 2.1–2.3	4	0.82	passed	4ceffd1cfb36

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
gum5-e3-molar-mass-glucose	JCGM GUM-5:2026, Example 3, Tables 3.1 and 3.2	2	0.20	passed	d7e7f39a7c87
gum5-e4-gravimetric-gas-mixture	JCGM GUM-5:2026, Example 4, Tables 4.1–4.6 and 4.3.6.9	19	0.79	passed	5461bf527cc5
gum5-e5-ghg-inventory-agriculture-shared-a	JCGM GUM-5:2026, Example 5, Tables 5.1–5.3 and 5.4.9	4	0.95	passed	ddced1756bf9
gum5-e5-ghg-inventory-agriculture	JCGM GUM-5:2026, Example 5, Tables 5.1–5.3 and 5.4.9	4	0.95	passed	afa1b4539bfb
iso13528-e1-censored-deleted	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.50	passed	d79653d9ec97
iso13528-e1-censored-half	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.85	passed	4a0e98d3bac8
iso13528-e1-censored-ignored	ISO 13528:2022, Annex E.1 (Table E.1)	2	0.18	passed	3c6267ae0c18
iso13528-e2-arsenic-homogeneity	ISO 13528:2022, E.2, Tables E.2 and E.3	8	0.65	passed	10f1f474c31c
iso13528-e3-atrazine-algorithm-a	ISO 13528:2022, E.3, Tables E.4 and E.5	13	0.84	passed	a5a9f01d4db0
iso13528-e3-atrazine-median	ISO 13528:2022, E.3, Table E.5	6	0.84	passed	646d4f79bdc4
iso13528-e4-mercury-feed	ISO 13528:2022, E.4, Tables E.6 and E.7	131	0.98	passed	f80274289f87
iso13528-e5-los-angeles-reference-value	ISO 13528:2022, Annex E.5 (Table E.8)	2	0.69	passed	fb8e1c1f2fa8
iso21748-c2-meat-content	ISO 21748:2017, Annex C.2, Table C.1	3	0.74	passed	e16b77cafad5
iso21748-c4-crude-fibre	ISO 21748:2017, Annex C.4, Tables C.6–C.9	6	0.99	passed	f7a71fc80945
iso5725-2-c1-sulfur-in-coal-reml	ISO 5725-2:2019, Annex C.1.9 (Table C.6), Annex B.2	16	0.91	passed	3401dfa58482
iso5725-2-c1-sulfur-in-coal	ISO 5725-2:2019, Annex C.1 (Tables C.1–C.5, Formulae C.1–C.24)	36	0.99	passed	62104ec9cf9d
iso5725-2-c3-creosote-oil-reml	ISO 5725-2:2019, Annex C.3.8 (Table C.19), Annex B.2	20	0.95	passed	20909b80d171
iso5725-2-c3-creosote-oil	ISO 5725-2:2019, Annex C.3 (Tables C.14–C.18) and 8.5 (Tables 1–4)	44	1.0	passed	fa32b0832d3c
iso5725-3-j1-carbon-steel	ISO 5725-3:2023, Annex J.1, Table J.1	1	0.13	passed	48e94c8cebe0
iso5725-3-j2-vanadium-steel	ISO 5725-3:2023, Annex J.2, Tables J.2–J.5	1	0.12	passed	5963d9ee245d
iso6143-b1-no-uv-linear	ISO 6143:2025, Annex B.2.2 (Tables B.1–B.6)	14	0.62	passed	144afd0e97e2
iso6143-b2-n2-natural-gas-covariances	ISO 6143:2025, Annex B.2.3 (Table B.10)	9	0.29	passed	5eeb4fbcfe0a
iso6143-b2-n2-natural-gas-linear	ISO 6143:2025, Annex B.2.3 (Tables B.7, B.8)	12	0.29	passed	15952e9c1ddd
iso6143-b2-n2-natural-gas-quadratic	ISO 6143:2025, Annex B.2.3 (Table B.9)	16	0.24	passed	35e7e507d216
iso6143-b3-no-ndir-exponential	ISO 6143:2025, Annex B.1 and B.2.4 (Tables B.11, B.12)	11	0.028	passed	de9178cb3724
iso6143-b3-no-ndir-power	ISO 6143:2025, Annex B.2.4 (Tables B.11, B.12)	14	0.64	passed	2f4905126e01
iupac2024-3a-argentometric-titration-lo	Possolo et al., Examples 3A and 4I; Meija et al., Eq. (19)	2	0.10	passed	23a70b037582
iupac2024-3e-4m-iridium-molar-mass-ayes	Possolo et al., Examples 3E, 4M, 4N	4	0.80	passed	83d35ae39d79
iupac2024-4a-water-hardness	Possolo et al., Examples 3C and 4A	2	0.89	passed	f26e42a6bf3a
iupac2024-4b-silicon-isotopic-abundances	Possolo et al., Example 4B	8	0.90	passed	020f0b992f1e
iupac2024-4f-ccqm-k145-nickel-possolo-lea	Possolo et al., Example 4F	4	0.92	passed	191f2fd219cf
iupac2024-4h-cadmium-standardization	Possolo et al., Example 4H	6	0.73	passed	676f526bdc77
iupac2024-4i-argentometric-titration	Possolo et al., Examples 3A and 4I	4	0.77	passed	1ac5d70808eb
iupac2024-5c-benzoic-acid-purity-bsta	Possolo et al., Example 5C	4	0.48	passed	7c79d40af518
mupr-b1-nitrate	Eurachem/CITAC Guide, Example B1	7	0.96	passed	14bd744fe2ea
mupr-b2-pooled-precision	Eurachem/CITAC Guide, Example B2, Tables B2.1–B2.3	6	0.50	passed	acf4f4c5f51a

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
mupr-b2-replicates-on-two-days	Eurachem/CITAC Guide, Example B2, Eq. (B2.8)–(B2.9)	1	0.051	passed	92b534629cc0
mupr-b3-linear-precision	Eurachem/CITAC Guide, Example B3, Table B3.1, Eq. (B3.1)	1	0.82	passed	1aac173149a7
mupr-b4-arsenic-one-crm	Eurachem/CITAC Guide, Example B4	9	0.98	passed	157dea4a0d3a
mupr-b5-crm-and-spiked-samples	Eurachem/CITAC Guide, Example B5	7	0.83	passed	0621d638a70c
mupr-b6-sediment-22-pt	Eurachem/CITAC Guide, Example B6, Tables B6.1–B6.3	11	0.93	passed	36b33cb67a77
mupr-ex2-reference-materials	Eurachem/CITAC Guide, Example 2	3	0.36	passed	d22128167278
mupr-ex3-spiked-samples	Eurachem/CITAC Guide, Example 3	2	0.39	passed	9d0fc8aa8b30
nicob-pcb28-sediment-degrees-of-freedom	NIST Consensus Builder user manual (Koepke, §4.2, Exhibits 6 and 7; §A.2, §A.4, §A.5.1	36	0.65	passed	4496e92490a8
nist-tn1900-e8-molecular-weight-s2	NIST Technical Note 1900, Example E8, Exhibit 12	5	0.52	passed	5eb8cd030d87
nist-tn1900-e9-cadmium-standard	NIST Technical Note 1900, Example E9	5	0.73	passed	461267d7bd72
nordtest-3-ammonium-pt	Nordtest TR 537 ed. 4 (2017), §3.2 and Appendix 4	7	0.94	passed	2cfd4e952b23
nordtest-5.3-ammonium-duplicates	Nordtest TR 537 ed. 4 (2017), §5.3 and Appendix 5	4	0.65	passed	44572cd9b0f3
nordtest-5.3-oxygen-duplicates	Nordtest TR 537 ed. 4 (2017), §5.3 and Appendix 6	2	0.93	passed	818066c0b900
nordtest-6.1-one-crm	Nordtest TR 537 ed. 4 (2017), §6.1	3	0.35	passed	fd25680700ef
nordtest-6.1-three-crms	Nordtest TR 537 ed. 4 (2017), §6.1	3	0.51	passed	7a90650eb056
nordtest-6.2-proficiency-tests	Nordtest TR 537 ed. 4 (2017), §6.2	3	0.71	passed	b0369555b563
nordtest-6.3-recovery	Nordtest TR 537 ed. 4 (2017), §6.3	3	0.57	passed	0ab762c2219d
nordtest-8-bod-crm	Nordtest TR 537 ed. 4 (2017), §8.2 Example A, Appendix 7	4	0.80	passed	5efed90d1d68
nordtest-8-bod-pt	Nordtest TR 537 ed. 4 (2017), §8.2 Example B, Table 6	5	0.97	passed	b9e9c4f2cff6
nordtest-8-pcb-crm	Nordtest TR 537 ed. 4 (2017), §8.3 Example C	3	0.76	passed	05ee0aca2326
nordtest-8-pcb-pt	Nordtest TR 537 ed. 4 (2017), §8.3 Example D	5	0.61	passed	52dacc1395be
palma2019-cr-marine-sediment	C. Palma, Table 9	2	0.11	passed	3bbc5811d4ea
palma2019-cu-marine-sediment	C. Palma, Table 9	2	0.75	passed	35f8f04d9226
palma2019-li-marine-sediment	C. Palma, Table 9	2	0.30	passed	8fa0b5f6e48a
palma2019-mn-marine-sediment	C. Palma, Table 9	2	0.22	passed	49607ffc6cd9
palma2019-ni-marine-sediment	C. Palma, Table 9	2	0.33	passed	11311487a5e7
palma2019-pb-marine-sediment	C. Palma, Table 9, Section 4.1	2	0.49	passed	f287368eec64
palma2019-zn-marine-sediment	C. Palma, Table 9	2	0.46	passed	48110fb13ded
possolo-bodnar-2018-chromium-isotope-ratio	Possolo, Bodnar, 2018, Tables 3 and 4	5	0.89	passed	0c4aefff8bfc
possolo-meija-2022-atomic-weight-lead	Possolo & Meija, Sums, Products, and Ratios: Atomic Weight of Lead, pp. 26–29	12	0.97	passed	19419876615a
possolo-meija-2022-plasma-anion-gap	Possolo & Meija, Sums, Products, and Ratios: Plasma Anion Gap, pp. 16–18	2	0.35	passed	491b7c71b971
quam-a1-cadmium-standard-volume	EURACHEM/CITAC Guide CG 4, Example A1, A1.4	2	0.59	passed	6e5e9c698534
quam-a1-cadmium-standard	EURACHEM/CITAC Guide CG 4, Example A1, Tables A1.2 and A1.3	3	0.38	passed	8a1981901836
quam-a2-naoh-standardisation-substance	EURACHEM/CITAC Guide CG 4, Example A2, A2.4	2	0.32	passed	88cc0255b152

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
quam-a2-naoh-standardisation	EURACHEM/CITAC Guide CG 4, Example A2, Tables A2.2 and A2.3	3	0.80	passed	267a7d47140a
quam-a3-hcl-titration	EURACHEM/CITAC Guide CG 4, Example A3, Tables A3.2 and A3.3	4	0.69	passed	0b7c591d630a
quam-a4-pesticides-combination	EURACHEM/CITAC Guide CG 4, Example A4, Tables A4.4 and A4.5	3	0.48	passed	c9de239e7d98
quam-a4-pesticides-topdown	EURACHEM/CITAC Guide CG 4, Example A4, Tables A4.2–A4.5	7	0.91	passed	1aabd2867018
quam-a5-cadmium-release	EURACHEM/CITAC Guide CG 4, Example A5, Tables A5.3 and A5.4	4	0.86	passed	ba9c48b7d745
quam-a6-crude-fibre	EURACHEM/CITAC Guide CG 4, Example A6, Tables A6.4–A6.5	6	0.43	passed	0ec5386edcd1
quam-a7-assay-standard	EURACHEM/CITAC Guide CG 4, Example A7, Eq. (8), Table A7.3	2	0.46	passed	a767a11a8280
quam-a7-lead-double-idms	EURACHEM/CITAC Guide CG 4, Example A7, Eq. (5), Tables A7.4–A7.8	2	0.16	passed	23920f968c3d
tr569-ch7-case1-statistical	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 1	4	0.72	passed	a010c1426afd
tr569-ch7-case2-target-mean	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 2	4	0.40	passed	7141d66120c1
tr569-ch7-case3-target-reference	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limits and the central line in X-charts, case 3	4	0	passed	474850bed722
tr569-ch7-range-case1-statistical	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limit in R or r%-charts, case 1	2	0.24	passed	77ba5783baaf
tr569-ch7-range-case2-target	Nordtest TR 569 ed. 6 (2026), Chapter 7, Setting the control limit in R or r%-charts, case 2	3	0.33	passed	20c2abbf6675
tr569-ex1-nickel-steel-xrf	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 1	5	0.52	passed	70b1c501730a
tr569-ex10-boron-toys-target	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 10	4	0	passed	5f87ef68ba91
tr569-ex12-pooling-sr-srw	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 12 and footnote 24	5	0.94	passed	a404e9ec7134
tr569-ex2-cobalt-steel-xrf	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 2	4	2.8×10^{-12}	passed	63859fea3d4d
tr569-ex3-ammonium-r	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 3	4	0.80	passed	e52d80ca7b97
tr569-ex3-ammonium-x	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 3	4	0.60	passed	00e3c1ff64dd
tr569-ex4-lead-water-icpms	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 4	4	1.1×10^{-13}	passed	dc2e963ef0d6
tr569-ex5-arsenic-crm-tolerance	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 5	2	0	passed	67a32af10def
tr569-ex6-phosphorus-relative-range	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 6	3	0.91	passed	ebfac2b11550
tr569-ex7-beta-hch-crm-target	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 7	5	0	passed	683e4066fa05
tr569-ex8-copper-r-review	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	6	0.76	passed	933d19ab852e
tr569-ex8-copper-x-new-limits	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	4	0.92	passed	9782b6d52166

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
tr569-ex8-copper-x-review	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 8	11	0.98	passed	ecec550b6284
tr569-ex9-zinc-procedural-blank	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 9	4	2.8×10^{-8}	passed	32389aea4393
tr569-table1-zinc	Nordtest TR 569 ed. 6 (2026), Chapter 2, Table 1 and Figure 1; Chapters 7 and 9	9	0.99	passed	9d4d05db4a66
ufs-a1-nitrate-lettuce	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A1, Table A1.1 and Figure A1.2	8	0.60	passed	b973c0f9827e
ufs-a2-lead-soil-uncertainty-factor	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A2, Tables A2.2–A2.4 and Figure A2.2	22	0.73	passed	d489ca48b672
ufs-a3-iron-groundwater-anova	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A3, Tables A3.5 and A3.7	3	0.42	passed	c66a94d4ddb1
ufs-a3-iron-groundwater-range	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A3, Tables A3.4 and A3.7	8	0.87	passed	469a337bd830
ufs-a4-vitamin-a-porridge-40g	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Tables A4.3, A4.5–A4.7, A4.14, A4.15 and summary	7	0.76	passed	0675d43f51af
ufs-a4-vitamin-a-porridge-4g	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Tables A4.4, A4.8–A4.10	5	0.80	passed	ae944956c424
ufs-a4-vitamin-a-qc-chart	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A4, Section 7.3, Table A4.13 and Figure A4.4; Section 13.2	5	0.86	passed	594e4283789d
ufs-a5-enzyme-feed-gy	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A5, Table A5.1; Section 10.2.7–10.2.8, Equations 6–8	8	0.96	passed	aef4031afda2
ufs-a6-cadmium-soil	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A6, Tables A6.1–A6.6	2	0.23	passed	58326ca9bce1
ufs-a6-phosphorus-soil	Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide UfS:2019, Appendix A, Example A6, Tables A6.1–A6.6	2	0.40	passed	9374b5c1c6b5
ume-procedure-19-2-qnmr-purity	TÜBİTAK UME Standard	4	0.76	passed	897f830365c3
ea402-s3-resistor-10kohm	EA-4/02 M:2022, S3	3	0.69	passed	1745db64f568
ea402-s6-power-sensor-18ghz	EA-4/02 M:2022, S6	4	0.71	passed	e6f4998f2942
ea402-s7-step-attenuator-30db	EA-4/02 M:2022, S7	3	0.94	passed	f467f5736cad
ea402-s9-dmm-100v	EA-4/02 M:2022, S9	2	0.85	passed	8974827a9eeb
gum-h2-impedance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.67	passed	18c8a39d92ed
gum-h2-reactance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.98	passed	2f502deaa633
gum-h2-resistance	JCGM 100:2008, H.2, Tables H.2 and H.3	2	0.34	passed	ecb068cde7db
jcgm101-9.4-comparison-loss-x01	JCGM 101:2008, 9.4, Table 9	6	0.11	passed	2603aca2f632
jcgm101-9.4-comparison-loss-x02	JCGM 101:2008, 9.4, Table 9	5	0.11	passed	083193ff79c7
jcgm101-9.4-comparison-loss-x03	JCGM 101:2008, 9.4, Table 8	6	0.042	passed	3f85dfd8181d
jcgm101-9.4-comparison-loss-x04	JCGM 101:2008, 9.4, Table 9	6	0.11	passed	5e8f1826fbae
jcgm101-9.4-comparison-loss-x05	JCGM 101:2008, 9.4, Table 9	5	0.11	passed	8d0ed6c5b6a3
jcgm101-9.4-comparison-loss-x06	JCGM 101:2008, 9.4, Table 8	6	0.39	passed	ab950243a37a
jcgm101-9.4-comparison-loss-x07	JCGM 101:2008, 9.4, Table 9	6	0.22	passed	d3dd9dc47dd2

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
jcgm101-9.4-comparison-loss-x001-correlated	JCGM 101:2008, 9.4, Table 9	5	0.22	passed	1e19c2357fce
jcgm101-9.4-comparison-loss-x001	JCGM 101:2008, 9.4, Table 8	6	0.99	passed	ff8f9bcb3688
jcgm102-9.3-polar-x001-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.43	passed	6f5e9329ca19
jcgm102-9.3-polar-x001	JCGM 102:2011, 9.3 (Table 6)	10	0.71	passed	3ca915fa8fa3
jcgm102-9.3-polar-x010-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.49	passed	5ce15f1a66b2
jcgm102-9.3-polar-x010	JCGM 102:2011, 9.3 (Table 6)	10	0.82	passed	4851a5122db2
jcgm102-9.3-polar-x100-correlated	JCGM 102:2011, 9.3 (Table 7)	10	0.81	passed	2faa0ee7654b
jcgm102-9.3-polar-x100	JCGM 102:2011, 9.3 (Table 6)	10	0.83	passed	693701192151
jcgm102-9.4-resistance-reactance-correlated	JCGM 102:2011, 9.4 (Tables 8–11)	17	0.98	passed	842b29ffc161
jcgm102-9.4-resistance-reactance	JCGM 102:2011, 9.4 (Tables 8–11)	17	0.98	passed	2b23311daaa3
m3003-c5-dvm-dominant-rectangular	UKAS M3003 ed. 6 (2024), Appendix C.5–C.10	4	0.55	passed	e56eb32e70b1
m3003-k1-resistor-10kohm	UKAS M3003 ed. 6 (2024), Appendix K.1	3	0.70	passed	36195ab2105a
m3003-k2-power-sensor-18ghz	UKAS M3003 ed. 6 (2024), Appendix K.2	3	1.0	passed	b17ec2b7bcc0
m3003-k3-attenuator-30db-mismatch-15	UKAS M3003 ed. 6 (2024), Appendix K.3 and E.9.5	3	1.7×10^{-11}	passed	692b9150fbde
m3003-k3-attenuator-30db	UKAS M3003 ed. 6 (2024), Appendix K.3	3	0.99	passed	b466a13f58f9
m3003-l3-dmm-1v-range-095v	UKAS M3003 ed. 6 (2024), Appendix L.3	3	0.53	passed	60e74c14aee6
m3003-l3-dmm-1v-range-1v-dominant	UKAS M3003 ed. 6 (2024), Appendix L.3	2	0.12	passed	097a8c15a007
m3003-l3-dmm-1v-range-500mv	UKAS M3003 ed. 6 (2024), Appendix L.3	2	0.51	passed	8b4de14783fb
nist-tn1900-e11-step-attenuator	NIST Technical Note 1900, Example E11, Exhibits 16 and 17	4	0.66	passed	22a47e8e8ccb
num-10-resistance	JCGM 100:2008, H.2, Tables H.2 and H.3	7	0.43	passed	9753964f0d92
possolo-meija-2022-wheatstone-bridge	Possolo & Meija, Sums, Products, and Ratios: Wheatstone Bridge, pp. 24–26	2	0.084	passed	1635e714f4a5
ume-procedure-19-1-dmm-10v-tubitak	TÜBİTAK UME, Section 19.1	4	0.35	passed	47c94ff7823f
cg19-ex1-flask-1000ml	EURAMET Calibration Guide No. 19 v4.1 (2025), 7.1 (Tables 1 and 2)	5	0.63	passed	1fbd3c1e8d4d
cg19-ex2-pycnometer-100ml	EURAMET Calibration Guide No. 19 v4.1 (2025), 7.2 (Tables 3 and 4)	5	0.85	passed	a0e2abe4e822
ea402-s12-water-meter-error	EA-4/02 M:2022, S12	2	0.59	passed	26d6ce337d68
ea402-s12-water-meter-mean-error	EA-4/02 M:2022, S12	3	0.74	passed	a2fdf31ca332
ea402-s12-water-meter-volume	EA-4/02 M:2022, S12	2	0.60	passed	e9b27f4329ab
m3003-k9-flowrate-standard-condition	UKAS M3003 ed. 6 (2024), Appendix K.9	4	0.90	passed	eb4e1f812383
nist-tn1900-e4-pitot-tube	NIST Technical Note 1900, Example E4	5	0.68	passed	bb91f86703b1
gum5-e9-torque-slope-ols	JCGM GUM-5:2026, Example 9, Tables 9.1–9.3	4	0.91	passed	d4e0c049019b
gum5-e9-torque-slope-wls	JCGM GUM-5:2026, Example 9, Tables 9.1–9.3	4	0.82	passed	e57e91bff780
iso21748-c1-co-emissions	ISO 21748:2017, Annex C.1	3	0	passed	ae10d6471462
iso21748-c3-aerobic-plate-counts-2l	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.54	passed	43a5bd2dbfa9
iso21748-c3-aerobic-plate-counts-2ml	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.85	passed	48f3292cd03a
iso21748-c3-aerobic-plate-counts-petrifilm	ISO 21748:2017, Annex C.3, Tables C.2–C.4	3	0.73	passed	dce6e6671de2
jcgm101-9.2.2-additive-gaussian	JCGM 101:2008, 9.2.2	7	0.38	passed	13a20e7c0733
jcgm101-9.2.3-additive-rectangular	JCGM 101:2008, 9.2.3	6	0.58	passed	1b6ba543e75d
jcgm101-9.2.4-additive-dominant-rectangular	JCGM 101:2008, 9.2.4	5	0.33	passed	e6df7f442a4b
jcgm102-9.2.2-additive-gaussian	JCGM 102:2011, 9.2.2 (Table 3)	16	0.45	passed	641460f72f9a
jcgm102-9.2.3-additive-rectangular	JCGM 102:2011, 9.2.3 (Table 4)	16	0.45	passed	9bbeb3090c69
jcgm102-9.2.4-additive-dominant-rectangular	JCGM 102:2011, 9.2.4 (Table 5)	16	0.56	passed	4893aab09c8d

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
jcgm106-7.3-bursting-strength	JCGM 106:2012, 7.3.3, Example 2	2	0.20	passed	8edb9ef79fbe
jcgm106-7.3-zener-diode	JCGM 106:2012, 7.3.3, Example 1	2	0.15	passed	5a8dceaa8573
jcgm106-7.4-motor-oil-viscosity	JCGM 106:2012, 7.4, Example	3	0.53	passed	eaff9c8aff4f
jcgm106-8.3-nandrolone	JCGM 106:2012, 8.3.3, Example 2	1	0.68	passed	1eb4125b4357
jcgm106-8.3-speed-limit	JCGM 106:2012, 8.3.3, Example 1	3	0.82	passed	3659745800c1
jcgm106-9.5-ball-bearings	JCGM 106:2012, 9.5.4, Example	5	0.76	passed	d7dcbb06ca16
jcgm106-9.5-precision-resistors	JCGM 106:2012, 9.5.3, Example	3	0.88	passed	7adcf527bba4
num-3-ratio-abc	T. Lafarge, §3, Figure 1	7	0.99	passed	ad65ac35ec61
tr569-ex11-micropipette-tolerance	Nordtest TR 569 ed. 6 (2026), Chapter 14, Example 11	2	0	passed	653424ca3244
tur-guard-band-dobbert-tur2-5	Sandia National Laboratories	6	3.7×10^{-9}	passed	f81949c27202
tur-guard-band-rp10-tur2-0	Sandia National Laboratories	6	3.9×10^{-9}	passed	4f4e9279235d
tur-guard-band-rss-tur1-5	Sandia National Laboratories	6	2.7×10^{-9}	passed	56485201c405
ea402-s10-vernier-calliper	EA-4/02 M:2022, S10	2	0.68	passed	ffc45534bcb4
ea402-s13-ring-gauge-90mm	EA-4/02 M:2022, S13	3	0.70	passed	4a6774391883
ea402-s13-ring-gauge-temperature	EA-4/02 M:2022, S13	2	0.40	passed	397ae464920a
ea402-s4-gauge-block-50mm	EA-4/02 M:2022, S4	3	0.92	passed	b3ea19f34d3d
gum-h1-end-gauge	JCGM 100:2008, H.1, Table H.1	5	0.83	passed	de2a2c7e90e5
jcgm101-9.5-gauge-block-approximate	JCGM 101:2008, 9.5, Tables 10 and 11	7	0.40	passed	310f23f32571
jcgm101-9.5-gauge-block-nonlinear	JCGM 101:2008, 9.5, Tables 10 and 11	7	0.51	passed	4492a781087f
m3003-k10-sheet-thickness	UKAS M3003 ed. 6 (2024), Appendix K.10	4	0.92	passed	7788b0ac7948
m3003-k6-gauge-block-10mm-temperature	UKAS M3003 ed. 6 (2024), Appendix K.6	3	0.068	passed	e16dd56d1329
m3003-k6-gauge-block-10mm	UKAS M3003 ed. 6 (2024), Appendix K.6	3	0.71	passed	85d98bbc4910
num-8-end-gauge-gaussian	T. Lafarge, §8, Table 3 and Figure 3	7	0.99	passed	34e9a89199b3
possolo-meija-2022-storage-tank-sums	Possolo & Meija, Sums, Products, and Ratios: Storage Tank, pp. 18–24	4	0.69	passed	f9ca3ca86723
cg18-h1-100g-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.97	passed	41f3b133f94f
cg18-h1-220g-max-option1-delta-L	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	5	0.65	passed	322a870fbac5
cg18-h1-220g-max-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.39	passed	7d380540f128
cg18-h1-220g-max-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	4	0.82	passed	5154c8d16e91
cg18-h1-220g-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1	5	0.77	passed	6dd4711c2234
cg18-h1-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1.1/A to H1.4/A; 7.4, 7.5; Appendix G	8	0.28	passed	fe1df4df2d7f
cg18-h1-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H1.1/B to H1.4/B; 7.4, 7.5; Appendix G	8	0.71	passed	d5dada5893c4
cg18-h2-60kg-10kg-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	5	0.96	passed	d897ae464801
cg18-h2-60kg-max-adjusted	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/B	5	0.96	passed	e5419b841ea0

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
cg18-h2-60kg-max-option1-delta-U	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.90	passed	621546a5a299
cg18-h2-60kg-max-option1	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	5	0.93	passed	990abcc77a74
cg18-h2-60kg-max-option2	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.89	passed	b74c636cad3d
cg18-h2-60kg-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2, H2.3/A	4	0.77	passed	d838031a9642
cg18-h2-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2.1/A to H2.4/A; 7.4, 7.5; Appendix G	19	0.30	passed	d5cb624eac61
cg18-h2-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H2.1/B to H2.4/B; 7.4, 7.5; Appendix G	19	0.47	passed	c3f7350afe3c
cg18-h3-30t-10t-weights	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	5	0.97	passed	2e8ca2acb3f3
cg18-h3-30t-15t-substitution	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	2	0.19	passed	f2d8228406ec
cg18-h3-30t-max-substitution	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	4	0.81	passed	e07e337fcebe
cg18-h3-30t-zero	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3, H3.2/A and H3.3/A	4	0.27	passed	cd9c318ba12d
cg18-h3-a-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3.1/A to H3.4/A; 7.4, 7.5; Appendix G	9	0.87	passed	e45f0947730d
cg18-h3-b-weighing-result	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H3.1/B to H3.4/B; 7.4, 7.5; Appendix G	9	0.67	passed	f4ae0f60adf2
cg18-h4-400g-error-function-sm0.25	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	14	0.97	passed	b5855bca1502
cg18-h4-400g-error-function-sm0.25	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	13	0.90	passed	2dda74a14c9e
cg18-h4-400g-error-function	EURAMET Calibration Guide No. 18 v4.0 (2015), Appendix H4; Appendix C2.2	4	0.81	passed	87200041a343
ea402-s2-weight-10kg	EA-4/02 M:2022, S2	3	0.87	passed	46f6088c84be
ilac-p14-cmc-floor-weight-10kg	EA-4/02 M:2022, S2	4	0.87	passed	69c5c7a99c61
jcgm101-9.3-mass-calibration-hybrid-mode2	JCGM 101:2008, 9.3, Table 6	6	0.73	passed	e8d4e7bc4aa4
jcgm101-9.3-mass-calibration	JCGM 101:2008, 9.3, Tables 6 and 7	7	0.97	passed	c6ddfed906ee
m3003-k4-weight-10kg	UKAS M3003 ed. 6 (2024), Appendix K.4	3	0.15	passed	ce1817c92061
m3003-k5-weighing-machine-200kg	UKAS M3003 ed. 6 (2024), Appendix K.5	3	0.50	passed	809b189d5416
nist-tn1900-e1-weighing	NIST Technical Note 1900, Example E1, Exhibit 1	7	0.27	passed	37b8767261d1
gum5-e14-radioactive-decay-bi207	JCGM GUM-5:2026, Example 14, 14.3 and Table 14.1	4	1.0	passed	12083ca095f7
gum5-e15-wire-rope-breaking-force	JCGM GUM-5:2026, Example 15, 15.3, 15.4 and 15.5	5	0.98	passed	5a3f63e1be03
iso5725-2-c2-softening-point-pitch-ref1	ISO 5725-2:2019, Annex C.2.9 (Table C.13), Annex B.2	16	0.91	passed	9bf797a3de20
iso5725-2-c2-softening-point-pitch	ISO 5725-2:2019, Annex C.2 (Tables C.7–C.12, Formulae C.25–C.33)	31	1.0	passed	5ba888152956
koepke2017-half-life-cs137	Koepke, §7.2, Tables 3 and 4	13	0.98	passed	e50d5cde68e1
koepke2017-half-life-sr90	Koepke, §7.2, Tables 3 and 4	13	0.61	passed	20802cda308f
koepke2017-newtonian-constant-G	Koepke, §7.1, Tables 1 and 2	20	0.94	passed	63a4273da38f

Project	Source	Values	max |Δ|/tol	Outcome	SHA-256
merkatas2019-newtonian-constant-gas	Merkatas, Tables 1 and 4	14	0.69	passed	a1aae532511e
nicob-co60-activity-leave-one-out	NIST Consensus Builder user manual (Koepke, §4.4, Exhibits 19, 21 and 22; §A.5.1	68	0.82	passed	c4640bb263c5
nist-tn1900-e13-thermal-expansion-gauge-95	NIST Technical Note 1900, Example E13, Exhibit 20	6	0.89	passed	05abdaaa5acc
nist-tn1900-e13-thermal-expansion-gauge-99	NIST Technical Note 1900, Example E13, Exhibit 20	6	0.89	passed	23c7b16dfc57
nist-tn1900-e13-thermal-expansion-steel-95	NIST Technical Note 1900, Example E13, Exhibit 20	4	0.86	passed	7a1013ee019a
nist-tn1900-e13-thermal-expansion-steel-99	NIST Technical Note 1900, Example E13, Exhibit 20	4	0.86	passed	decd6bb40fcc
nist-tn1900-e3-falling-ball-viscometer	NIST Technical Note 1900, Example E3, Exhibit 3	7	0.55	passed	f4d03a720999
num-11-stefan-boltzmann	T. Lafarge, §11, Table 5 and Figure 7	6	0.69	passed	28def2666dab
possolo-meija-2022-hubble-lemaitre-constant	Possolo & Meija, Consensus Building, Hubble-Lemaître Constant, pp. 146–151	3	0.36	passed	6be95d5d53cd
emue-e2.2-vacuum-gauge-correction-applied	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.3	4	0.90	passed	19e0b54745a0
emue-e2.2-vacuum-gauge-correction-ignored	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.7	4	0.92	passed	1f42fa6f91e2
emue-e2.2-vacuum-gauge-correction-uncertainty	EMPIR 17NRM05 EMUE, Example E2.2, Tables E2.2.1–E2.2.5	4	0.92	passed	5a5fd2d10239
emue-e2.2-vacuum-gauge-sum-model-corrected	EMPIR 17NRM05 EMUE, Example E2.2, E2.2.7	4	0.068	passed	861d1957021f
emue-e2.2-vacuum-gauge-sum-model-ignored	EMPIR 17NRM05 EMUE, Example E2.2, E2.2.7	4	0.013	passed	08bfc3dacf84
m3003-k8-pressure-indicator-2mpa	UKAS M3003 ed. 6 (2024), Appendix K.8	3	0.45	passed	b24083f194e4
ea402-s11-block-calibrator-180c	EA-4/02 M:2022, S11	2	0.58	passed	623d8617c638
ea402-s5-thermocouple-emf	EA-4/02 M:2022, S5	3	0.67	passed	8f1a7a902650
ea402-s5-thermocouple-furnace	EA-4/02 M:2022, S5	3	0.37	passed	63cb7c26ccea
jcgm102-9.5.2-resistance-thermometer	JCGM 102:2011, 9.5.2 (Tables 12, 13)	2	0.59	passed	be62be046ce0
jcgm102-9.5.3-ten-temperatures	JCGM 102:2011, 9.5.3 (Tables 14–16)	31	0.96	passed	d48d618512c8
m3003-k11-gas-temperature	UKAS M3003 ed. 6 (2024), Appendix K.11 (K.11.17–K.11.18)	3	0.75	passed	fba975668f0d
m3003-k11-probe-calibration	UKAS M3003 ed. 6 (2024), Appendix K.11 (K.11.9)	2	0.51	passed	5d2049923f16
m3003-k7-thermocouple-emf	UKAS M3003 ed. 6 (2024), Appendix K.7	3	0.75	passed	d1eb52c4815e
m3003-k7-thermocouple-furnace	UKAS M3003 ed. 6 (2024), Appendix K.7 (K.7.19)	2	0.26	passed	0e0a20de7c1f
nist-tn1900-e2-surface-temperature	NIST Technical Note 1900, Example E2, Exhibit 2	6	0.98	passed	f51b0acd8aa8

The column max |Δ|/tol gives the largest ratio of the difference between computed and reference value to its tolerance in the file (≤ 1 passes). The tolerances reflect the rounding of the published values or, for Monte Carlo results, about five times the measured dispersion (research log D15).

Checked by: _____ Date: _____ Signature: _____

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